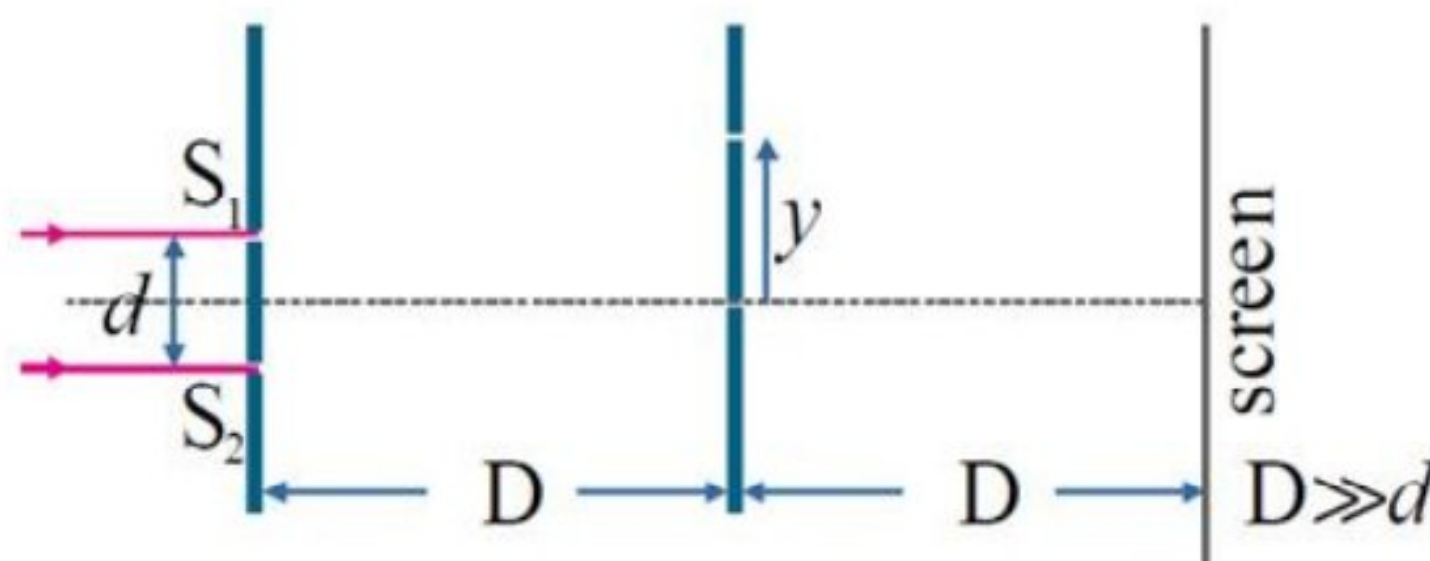


Single Choice Questions :

- In a Young's double slit experiment the slit separation is 0.5 mm and the screen is 0.5 m from the slit. For a monochromatic light of wavelength 500 nm the distance of 3rd maxima from the 2nd minima on the other side is :
(A) 2.75 mm (B) 2.5 mm (C) 22.5 mm (D) 2.25 mm
- In Young's double slit experiment how many maxima can be obtained on a screen (including the central maximum) on both sides of the central fringe if ?
 $\lambda = 200 \text{ nm}$ and $d = 700 \text{ nm}$.
(A) 12 (B) 7 (C) 18 (D) 4
- In a YDSE with identical slits, the intensity of the central bright fringe is I_0 . If one of the slits is covered, the intensity at the same point is:
(A) $2I_0$ (B) I_0 (C) $I_0/2$ (D) $I_0/4$
- At a certain point on a screen in a YDSE apparatus the path difference for the two interfering rays is $(1/3)\text{rd}$ of the wavelength. Find the ratio of the intensity at this point to that at the centre of a bright fringe.
(A) $1/2$ (B) $1/4$ (C) $1/3$ (D) $3/4$
- The ratio of the intensity at the centre of a bright fringe to the intensity at a point one-quarter of the fringe width from the centre is :
(A) 2 (B) $1/2$ (C) 4 (D) 16
- In the Young's double slit experiment apparatus shown in figure, the ratio of maximum to minimum intensity on the screen is 9. The wavelength of light used is λ , then the value of y is :



- (A) $\frac{\lambda D}{d}$ (B) $\frac{\lambda D}{2d}$ (C) $\frac{\lambda D}{3d}$ (D) $\frac{\lambda D}{4d}$